Notes: Tintelnot (QJE, 2017)

Global Production with Export Platforms

Contents

1 Big picture

- **Question.** How should we model multinational firms that choose where to locate production plants, how much to produce in each plant, and which foreign markets to serve from each plant?
- **Core empirical fact.** Foreign affiliates of multinationals do not only serve host-country markets. They also export a large share of output to third countries. These export-platform sales account for a large fraction of multinational production.
- **Core modeling challenge.** A multinational with fixed costs of opening plants and the ability to serve many markets from any plant faces a difficult combinatorial problem: choose plant locations and allocate product-market production across those locations.
- **Main modeling contribution.** The paper makes this problem tractable by assuming each firm produces a continuum of products and draws product-location productivities from Frechet distributions. This makes firm-level revenues smooth functions of plant sets and cost parameters.
- **Key economic frictions.** The model includes three barriers: iceberg trade costs, variable efficiency losses from foreign production, and fixed costs of establishing foreign plants.
- **Estimation.** The paper uses German firm-level data from the MiDi database to estimate foreign production costs and fixed costs of foreign plant entry. Output across locations identifies variable production costs; observed plant locations identify fixed costs.
- **Calibration.** The estimated firm-level parameters are embedded in a general equilibrium model for 12 Western European and North American countries using trade and multinational production shares.
- **Main quantitative findings.** Fixed costs of foreign investment are large, export platforms are quantitatively important, and ignoring export platforms or fixed costs changes counterfactual predictions.
- **Main counterfactual 1.** A Canada-EU trade and investment agreement would substantially redirect EU multinational production from the United States toward Canada in the full global production model.
- **Main counterfactual 2.** U.S. technology improvements generate much larger welfare gains abroad when multinational production is allowed because U.S. firms use better technology in foreign plants.
- **Main takeaway.** Multinational production is not just a substitute for trade. It is a technology-transmission and production-location channel whose effects depend critically on export platforms and fixed costs.

2 Incremental contribution of the paper

2.1 Contribution relative to trade-only gravity models

- Eaton-Kortum-style trade models explain how countries source goods internationally, but firms remain country-based and do not choose foreign production locations.
- Tintelnot embeds an Eaton-Kortum structure *inside the firm*: each firm has a continuum of products, and each plant receives product-specific productivity draws.
- This allows the model to generate closed-form firm sales and production shares by location even when firms operate multiple plants.

Interpretation: The incremental step is to move from country-level sourcing to firm-level global production networks, while preserving tractability.

2.2 Contribution relative to multinational-production models

- Earlier quantitative multinational-production models often ruled out export-platform sales or ignored fixed costs of foreign investment.
- The paper allows both: affiliates may export to third countries, and firms pay fixed costs to establish foreign operations.
- This combination is essential because plant-location choices and export-platform sales are jointly determined.

Interpretation: The model captures a central empirical feature of multinational firms: a plant in a foreign country can be an export platform rather than merely a local-market affiliate.

2.3 Contribution relative to empirical work on FDI costs

- The paper estimates both variable foreign production costs and fixed plant-entry costs using firm-level multinational data.
- Variable costs are identified from within-firm comparisons of output across countries.
- Fixed costs are identified from which countries firms choose to enter, conditional on productivity and variable costs.
- This is richer than using only aggregate bilateral multinational production flows.

Interpretation: The paper shows how to use detailed firm-location data to separately identify variable and fixed barriers to multinational production.

2.4 Contribution to policy counterfactuals

- The model implies that trade and investment agreements can redirect multinational production across third countries.
- Models without export platforms miss this relocation channel.
- Models without fixed costs overstate how freely multinationals can reconfigure plant networks.

Interpretation: The paper's policy contribution is to show that counterfactuals involving investment liberalization require a model with both fixed investment costs and export-platform sales.

3 Model and main equations

3.1 Demand

Consumers in country j have CES preferences over firms and over each firm's continuum of varieties. The demand for variety υ from firm ω is

$$q_j(\omega, \upsilon) = p_j(\omega, \upsilon)^{-\sigma} \frac{Y_j}{P_j^{1-\sigma}}, \tag{1}$$

where Y_j is income and P_j is the aggregate price index.

The firm-level price index in market j is

$$p_j(\omega) = \left(\int_0^1 p_j(\omega, \upsilon)^{1-\sigma} d\upsilon \right)^{1/(1-\sigma)}, \tag{2}$$

and firm-level expenditure in market j is

$$s_j(\omega) = p_j(\omega)^{1-\sigma} \frac{Y_j}{P_j^{1-\sigma}}. \tag{3}$$

Interpretation: This is standard CES demand, except each firm supplies a continuum of products. The continuum assumption smooths the firm's production-location problem.

3.2 Firm productivity and plant productivity

A firm from home country i is characterized by:

- core productivity ϕ;
- a set of production locations Z;
- country-specific fixed costs η;
- plant-specific productivity shifters Φ_j.

For a product produced in plant country j, productivity ν_j follows a Frechet distribution:

$$\Pr(\nu_j \leq x) = \exp\left[-\phi^\theta \Phi_j^\theta \gamma_{ij}^{-\theta} x^{-\theta}\right]. \tag{4}$$

- γ_{ij} is the variable efficiency loss of producing in country j for a firm from country i.
- Larger $\phi\Phi_j$ means better productivity draws.
- A lower θ means more dispersion in product-level productivity.

Interpretation: The Frechet structure lets the firm allocate different products to different plants while obtaining closed-form shares, just as in Eaton and Kortum.

3.3 Production and sourcing after plant choice

Given plant set Z, wage w_l, and trade cost τ_{lm}, the cost of serving market m from plant l depends on $w_l\tau_{lm}$ and productivity. The firm serves each product-market from the cheapest plant.

The distribution of product-level costs in market m is

$$G_m(c \mid i, \phi, Z, \Phi) = 1 - \exp\left[-\sum_{k\in Z}\left(\frac{\phi\Phi_k}{\gamma_{ik}w_k\tau_{km}}\right)^\theta c^\theta\right]. \tag{5}$$

This yields a firm-level price index in market m:

$$p_m(i, \phi, Z, \Phi) = \kappa\phi^{-1}\left[\sum_{k\in Z}(\gamma_{ik}w_k\tau_{km})^{-\theta}\Phi_k^\theta\right]^{-1/\theta}, \tag{6}$$

where κ is a constant that depends on θ and σ.

Interpretation: A plant is valuable if it has high plant productivity, low wages, low foreign production losses, and low shipping costs to important markets.

3.4 Plant-location choice

Before observing plant-specific productivity shocks, the firm chooses Z to maximize expected profits net of fixed costs:

$$E_\Phi\left[\Pi(i,\phi,Z,\Phi)\right] - \sum_{k\in Z, k\neq i} \eta_k. \tag{7}$$

The location choice probability is

$$\Pr(Z = Z_t \mid \phi_t) = \Pr\left(E_\Phi \Pi(\phi_t, Z_t, \Phi, \eta) \geq E_\Phi \Pi(\phi_t, Z', \Phi, \eta)\ \forall Z'\right). \tag{8}$$

Interpretation: Fixed costs create sparse plant networks. Export platforms matter because each plant's value depends on all destination markets, not only on host-country demand.

3.5 Likelihood and estimation logic

Given a firm's observed plant set Z_t and its revenues r_t in each plant, the likelihood combines:

1. the probability of choosing plant set Z_t;
2. the density of observed plant-level revenues conditional on Z_t.

The likelihood is written schematically as

$$L(\Theta) = \prod_t \int \Pr(Z = Z_t \mid \phi; \Theta)\, g(r_t \mid Z_t, \phi; \Theta) dG(\phi; \Theta). \tag{9}$$

Interpretation: Output by plant identifies productivity shifters conditional on parameters. Plant-entry choices identify fixed costs.

4 Data and calibration

4.1 German multinational firm data

- The firm-level estimation uses the German Microdatabase Direct Investment, MiDi.
- German resident investors are legally required to report foreign affiliates above voting-rights and balance-sheet thresholds.
- The paper uses manufacturing-sector German multinationals and their majority-owned foreign affiliates in 12 Western European and North American countries.
- The estimation observes the set of foreign countries in which a German firm has plants and the firm's output in each country.

Interpretation: This is exactly the type of data needed for the model: plant locations and plant-level output by firm.

4.2 Country-level and aggregate data

- The calibration uses bilateral trade shares, multinational production shares, wages, GDP, and price indices.
- The model is calibrated for 12 Western European and North American countries.
- Trade costs are disciplined by distance, language, and contiguity.
- Multinational production costs are disciplined by bilateral production shares and firm-level estimates.

Interpretation: The firm-level estimates discipline micro barriers, while the aggregate calibration embeds them in general equilibrium.

5 Empirical and quantitative results

5.1 Estimated production and fixed costs

- Variable unit input costs for German firms are lowest in nearby countries such as Austria and highest in the United States.
- Estimated fixed costs of foreign affiliates are large, measured in millions of euros.
- Fixed costs vary substantially across destinations.
- The model reproduces the average share of output produced abroad by German multinationals.

Interpretation: Multinational expansion is constrained by both variable efficiency losses and large plant-entry fixed costs. This explains why firms operate sparse global production networks.

5.2 Home bias in production

The paper asks why German multinationals keep so much output at home. Table II decomposes the sources of home bias.

- In the data, the average foreign production share is 0.288.
- The estimated model predicts 0.278.
- If foreign unit input costs were the same as in Germany, the foreign share would rise to 0.566.
- Removing fixed costs raises the foreign share to 0.708.
- Removing both fixed costs and foreign unit-cost disadvantages raises the foreign share to 0.874.

Interpretation: Both variable foreign production losses and fixed costs are quantitatively important for home bias.

5.3 Calibration and export platform fit

- The global production model fits trade and multinational production moments.
- It also predicts export platform shares well relative to U.S. BEA data.
- A model without export-platform sales mechanically predicts zero export-platform shares.
- A no-fixed-cost model predicts positive export-platform shares but misses the sparse plant-network mechanism.

Interpretation: Export platform behavior is not a minor detail. It is essential for matching how multinationals actually use foreign affiliates.

5.4 Counterfactual: Canada-EU trade and investment agreement

The paper lowers both variable and fixed multinational production costs between Canada and EU countries by 20 percent.

- The global production model predicts substantial relocation of EU multinational production toward Canada.

- EU inward production shares in Canada rise sharply, while EU production in the United States falls.
- Models without export platforms predict much smaller effects on third-country production.
- Welfare gains are largest for Canada and positive but moderate for EU countries, while the United States loses slightly.

Interpretation: Trade and investment agreements affect not only bilateral production but also the geography of third-country export-platform production.

5.5 Counterfactual: U.S. technology improvement

The paper increases the productivity of all U.S.-origin firms by 20 percent.

- In a pure trade model, foreign welfare gains from U.S. technology decay sharply with distance and are small.
- In the global production model, foreign countries gain much more because U.S. multinationals operate foreign plants with improved technology.
- Canada, Ireland, and other countries with large U.S. multinational presence gain particularly strongly.
- The United States itself gains more with multinational production because its firms can use improved technology globally and earn higher profits.

Interpretation: Multinational production is a technology diffusion channel. It transmits firm-origin productivity improvements to production in host countries.

6 Identification and empirical design

6.1 Identification of variable production costs

- The key identifying variation is within-firm output across multiple plant countries.
- Core productivity is held fixed within firm.
- If a German firm produces relatively little in a foreign country, conditional on market potential and trade costs, this indicates higher foreign production cost or lower productivity there.

Interpretation: The model uses within-firm comparisons to distinguish country-specific foreign production costs from firm productivity.

6.2 Identification of fixed costs

- Once variable costs and firm productivities are disciplined, the location choices reveal fixed costs.
- Countries with many plants have lower average fixed costs.
- The relationship between firm productivity and the number of plants helps identify fixed-cost dispersion.

Interpretation: Fixed costs are inferred from who enters which countries, not from plant output conditional on entry.

6.3 Limits and assumptions

- The model abstracts from firm-level fixed costs of exporting.
- Productivity shocks are learned after plant-entry decisions, which helps tractability and estimation.
- The sample uses German manufacturing multinationals for firm-level estimation, then calibrates a broader multicountry system.
- Counterfactuals depend on the assumed trade-cost and multinational-cost structure.

Interpretation: The paper is a tractable quantitative framework, not a complete model of every margin of multinational activity. Its main strength is joint treatment of export platforms and fixed plant-entry costs.

7 Tables and figures

7.1 Figure I and Table I: export platforms and German multinational estimates

Read:

- Figure I shows that U.S. multinational affiliates in Europe export a large share of host-country output to third countries.
- Smaller European countries often have higher export-platform shares.
- Table I reports maximum likelihood estimates from German firm-level data.
- Estimated fixed costs are large, and unit input costs vary substantially across host countries.
- The model also reports country-level data moments such as the number of German firms, mean output, and median output.

Economic message: The paper's motivation and estimation are linked: export platforms are quantitatively important, and firm-level data reveal large fixed and variable costs of foreign production.

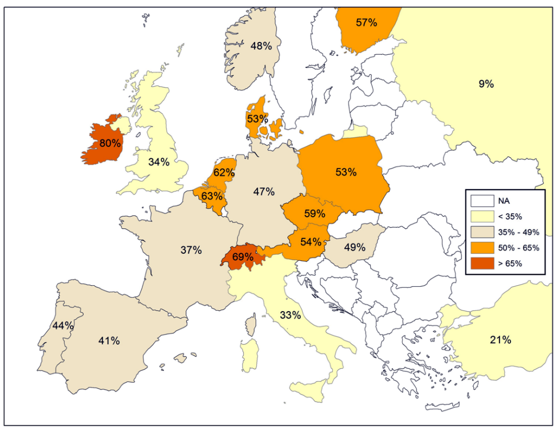

FIGURE I

Export Platform Shares for U.S. Multinationals in Europe

This figure displays the share of output that is exported to countries outside the host country by U.S. multinationals' majority-owned foreign affiliates in the manufacturing sector. For the European countries displayed in the figure, typically only about 5% of the output is sold back to the United States. An exception is Ireland, for which 17% of the output is sold to the United States. Across all countries (including countries outside Europe), U.S.-owned foreign affiliates in the manufacturing sector sell 43% of their output to countries outside the host country – 13% of the output is sold to the United States and 30% to other foreign countries. The statistics are for the year 2004. *Source:* BEA.

Figure I: export platform shares

TABLE I

MAXIMUM LIKELIHOOD ESTIMATES, IMPLIED FIXED COSTS, AND DESCRIPTIVE STATISTICS

	Estimated			Data		
Country	Unit input costs $\tilde{w}$	Fixed costs $\mu_{\tilde{\eta}}$	Mean fixed costs of established affiliates	Number of firms	Mean output	Median output
Austria	1.051	3.544	6.796	91	76.3	34
	(0.022)	(0.235)	(0.779)			
Belgium	1.150	3.859	9.904	45	235.3	37
	(0.054)	(0.278)	(2.768)			
Canada	1.261	3.776	8.755	36	536.0	28.5
	(0.043)	(0.222)	(1.761)			
Switzerland	1.155	3.462	6.050	70	58.3	17
	(0.023)	(0.243)	(1.156)			
Spain	1.155	3.207	7.632	117	191.9	32
	(0.019)	(0.229)	(1.198)			
France	1.152	3.227	7.287	191	107.7	30
	(0.015)	(0.179)	(0.729)			
United Kingdom	1.234	3.176	7.204	121	119.4	23
	(0.016)	(0.234)	(1.096)			
Ireland	1.106	4.054	5.438	18	36.3	19.5
	(0.045)	(0.323)	(1.299)			
Italy	1.235	3.284	7.072	100	65.0	27.5
	(0.023)	(0.221)	(0.972)			
Netherlands	1.118	3.724	7.927	46	83.1	25
	(0.024)	(0.246)	(1.650)			
United States	1.353	3.205	6.798	211	569.0	26
	(0.023)	(0.175)	(0.825)			

Std. dev. log fixed cost, $\sigma_{\tilde{\eta}}$	1.086	Scale parameter productivity, μ_ϕ	0.783
	(0.107)		(0.003
Shape parameter productivity, σ_ϕ	6.436	Std. dev. log productivity shock, σ_ϵ	0.108
	(0.313)		(0.005
Log-likelihood	−9.86E+03	Number of firms, T	9288

Table I: firm-level MLE estimates

7.2 Tables II and III: home bias decomposition and calibration

Read:

- Table II decomposes the average foreign production share of German multinationals.
- The estimated model nearly matches the data: 0.278 versus 0.288.
- Equalizing unit input costs with Germany raises foreign production to 0.566.
- Eliminating fixed costs raises it to 0.708.
- Table III reports calibrated trade and multinational production parameters for the pure trade, global production, no-export-platform, and no-fixed-cost models.

Economic message: Home bias in production is not explained by one friction. Both variable foreign production costs and fixed costs materially limit foreign production.

TABLE II

AVERAGE SHARE OF FOREIGN PRODUCTION IN THE OUTPUT OF GERMAN MULTINATIONALS

ata	Model	Same unit input costs as in Germany	No fixed costs	No fixed costs and same unit input costs as in Germany
288	0.278	0.566	0.708	0.874
	(0.010)	(0.023)	(0.010)	(0.003)

Table II: home-bias decomposition

TABLE III

CALIBRATED PARAMETERS

	Pure trade model	Global production model	No export platform model	No fixed costs model
Trade cost				
constant, β^{τ}_{const}	0.723	0.781	0.940	0.796
distance, β^{τ}_{dist}	0.139	0.121	0.083	0.115
language, β^{τ}_{lang}	0.922	0.926	0.898	0.923
contiguity, β^{τ}_{contig}	0.934	0.931	0.906	0.937
Variable MP cost				
constant, β^{γ}_{const}		1.211	0.958	1.974
distance, β^{γ}_{dist}		0.004	0.028	0.015
language, β^{γ}_{lang}		0.984	0.964	0.988
contiguity, β^{γ}_{contig}		0.944	0.936	0.867
Fixed MP cost				
constant, β^{η}_{const}		2.608	2.059	
distance, β^{η}_{dist}		0.000	0.000	
language, β^{η}_{lang}		0.851	0.186	
contiguity, β^{η}_{contig}		1.429	1.455	
dispersion, β^{η}_{disp}		0.262	1.091	
Norm trade fit	0.258	0.242	0.224	0.221
Norm MP fit		0.172	0.250	0.318

model the trade elasticity is distinct but close (−5.38). That the

Table III: calibrated parameters

7.3 Tables IV and V: export-platform fit and Canada-EU counterfactual

Read:

- Table IV compares BEA export-platform shares with model predictions.
- The global production model fits export-platform shares reasonably well.
- Table V reports the counterfactual effect of lower EU-Canada multinational production costs on inward multinational production shares.
- The global production model predicts a large increase in EU production in Canada and a decline in EU production in the United States.

Economic message: Export platforms are essential for predicting third-country relocation effects of investment agreements.

EXPORT PLATFORM SHARES: DATA AND MODELS

	Data	Global production model	No fixed costs model
Austria	0.54	0.49	0.46
Belgium	0.63	0.76	0.74
Canada	0.37	0.30	0.23
Switzerland	0.69	0.70	0.68
Germany	0.47	0.21	0.16
Spain	0.41	0.13	0.10
France	0.37	0.24	0.19
United Kingdom	0.34	0.20	0.15
Ireland	0.80	0.66	0.63
Italy	0.33	0.12	0.09
Netherlands	0.62	0.52	0.48

Table IV: export-platform shares

TABLE V

COUNTERFACTUAL CHANGES OF LOWER EU-CANADA MP COSTS: DIFFERENCE IN INWARD MP SHARES

	Global production model		No export platform model		No fixed costs model	
	Canada	United States	Canada	United States	Canada	United States
anada	−7.51	0.01	−2.00	0.01	−6.43	0.00
U countries	11.51	−0.24	3.64	0.01	7.68	−0.02
witzerland	−0.11	0.00	−0.02	0.00	−0.02	0.00
nited States	−3.89	0.23	−1.62	−0.02	−1.23	0.01

Notes. Counterfactual: reduction in variable and fixed MP costs between EU and Canada by 20%. Differ-

Table V: EU-Canada MP cost counterfactual

7.4 Tables VI and VII: technology and gains from multinational production

Read:

- Table VI reports welfare gains from a 20 percent productivity improvement for U.S. firms.
- Foreign gains are much larger in the global production model than in the pure trade model.
- Table VII reports gains from multinational production: welfare and real wage effects are positive in most countries, but real profit effects can be negative.
- The global production model produces different implications from no-export-platform and no-fixed-cost alternatives.

Economic message: Multinationals are a technology-transfer channel. They also lower price indices and raise real wages, while changing profit shares and market shares across countries.

TABLE VI

GAINS FROM U.S. TECHNOLOGY IMPROVEMENT

	Pure trade model	Global production model	No export platform model	No fixed costs model
Austria	1.0009	1.0299	1.0033	1.0210
Belgium	1.0005	1.0199	0.9806	1.0126
Canada	1.0070	1.0410	1.0551	1.0225
Switzerland	1.0007	1.0231	0.9802	1.0176
Germany	1.0003	1.0066	1.0038	0.9994
Spain	1.0005	1.0195	1.0172	1.0041
France	1.0003	1.0092	1.0029	1.0004
United Kingdom	1.0006	1.0149	1.0282	1.0016
Ireland	1.0022	1.0505	1.0348	1.0348
Italy	1.0004	1.0133	1.0118	1.0013
Netherlands	1.0006	1.0219	1.0089	1.0106
United States	1.1987	1.2222	1.2133	1.2138

Table VI: gains from U.S. technology improvement

TABLE VII

GAINS FROM MULTINATIONAL PRODUCTION

	Global production model			No export platform model			No fixed costs model		
	Welfare change	R. profit change	R. wage change	Welfare change	R. profit change	R. wage change	Welfare change	R. profit change	R. wage change
ustria	1.038	0.733	1.099	1.053	0.914	1.080	1.046	0.738	1.108
elgium	1.027	0.761	1.080	1.044	1.084	1.036	1.028	0.784	1.077
anada	1.029	0.806	1.074	1.094	0.973	1.118	1.022	0.904	1.046
witzerland	1.032	0.740	1.091	1.056	1.087	1.049	1.042	0.719	1.107
ermany	1.013	0.924	1.030	1.019	0.923	1.038	1.015	1.036	1.011
pain	1.020	0.844	1.056	1.027	0.842	1.064	1.016	0.954	1.029
rance	1.015	0.900	1.038	1.024	0.916	1.046	1.018	1.015	1.019
nited Kingdom	1.018	0.875	1.047	1.050	0.886	1.083	1.011	0.995	1.015
reland	1.044	0.682	1.117	1.067	0.940	1.093	1.060	0.619	1.148
aly	1.017	0.879	1.044	1.022	0.867	1.053	1.016	0.997	1.020
etherlands	1.025	0.793	1.071	1.028	0.902	1.053	1.024	0.860	1.057
nited States	1.008	1.003	1.009	1.010	0.992	1.014	1.012	1.066	1.001

Notes. Counterfactual: prohibitive costs of multinational production. A number in this table represents the outcome from the calibrated model divided by the outcome from the ıme model with no multinational production.

Table VII: gains from multinational production

7.5 Appendix Tables A.1 and A.2: foreign production shares

Read:

- Appendix Table A.1 shows that firms with more production locations produce a larger share abroad.
- Appendix Table A.2 shows foreign production shares by sector.
- These appendix tables document the firm-level and sector-level moments underlying the estimation.

Economic message: The foreign production share rises with the breadth of a firm's plant network, consistent with the importance of fixed costs and firm productivity for multinational scope.

APPENDIX TABLE A.1

FOREIGN PRODUCTION SHARES BY NUMBER OF PRODUCTION LOCATIONS

Number of production locations	Number of firms	Mean share of foreign production	Mean share of foreign production potential
2	474	0.26 (0.20)	0.37 (0.24)
3	102	0.32 (0.18)	0.54 (0.19)
4	40	0.35 (0.19)	0.65 (0.13)
5	23	0.39 (0.16)	0.71 (0.10)
6	14	0.46 (0.15)	0.75 (0.08)
$\geqslant 7$	12	0.48 (0.06)	0.80 (0.07)
All	665	0.29 (0.20)	0.44 (0.25)

Notes. Statistics for German MNE activities in 12 Western European and North American countries. Standard deviations in parentheses. *Source:* MiDi database.

Appendix Table A.1: production locations

APPENDIX TABLE A.2

FOREIGN PRODUCTION SHARES BY SECTOR

Sector	Number of firms	Mean number of production locations	Mean share of foreign production	Mean share of foreign production potential
Manufacture of ...				
textiles	15	2.27 (0.80)	0.34 (0.22)	0.39 (0.25)
publishing, printing, and reproduction of recorded media	22	2.36 (0.66)	0.26 (0.25)	0.37 (0.23)
chemicals and chemical products	85	3.05 (1.79)	0.33 (0.22)	0.45 (0.26)
rubber and plastic products	67	2.73 (1.21)	0.32 (0.21)	0.45 (0.25)
other nonmetallic mineral products	23	2.65 (1.19)	0.39 (0.24)	0.34 (0.21)
basic metals	31	2.35 (0.66)	0.22 (0.14)	0.40 (0.24)
metal products	72	2.32 (0.78)	0.27 (0.17)	0.43 (0.23)
machinery and equipment	138	2.49 (1.16)	0.25 (0.17)	0.46 (0.26)
electrical machinery and apparatus	34	2.79 (1.65)	0.26 (0.17)	0.48 (0.26)
radio, television, and communication equipment and apparatus	15	2.33 (0.72)	0.24 (0.16)	0.51 (0.28)
medical, precision, and optical instruments, watches, and clocks	49	2.33 (0.75)	0.30 (0.20)	0.54 (0.24)
motor vehicles, trailers and semi-trailers	57	2.82 (1.28)	0.30 (0.21)	0.48 (0.25)
All	665	2.57 (1.20)	0.29 (0.20)	0.44 (0.25)

Notes. Statistics for German MNE activities in 12 Western European and North American countries. Standard deviations in parentheses. Statistics are displayed for sectors with more than 10 German multinationals. *Source:* MiDi database.

Appendix Table A.2: sectors

8 Short summary

- The paper develops a tractable multicountry model of multinational firms with export-platform sales and fixed costs of foreign investment.
- The key modeling move is to put an Eaton-Kortum product-location productivity structure inside each firm.
- German firm-level data identify variable foreign production costs from output by location and fixed costs from plant-location choices.
- Fixed costs of foreign plants are large, and variable foreign production costs vary substantially across destinations.
- The model matches German multinationals' average foreign production share and fits BEA export-platform shares reasonably well.
- Removing fixed costs or equalizing foreign unit costs substantially raises predicted foreign production, showing that both frictions matter.
- A Canada-EU agreement would reallocate EU multinational production toward Canada and away from the United States.
- U.S. technology improvements produce much larger foreign welfare gains in the global production model than in a pure trade model.
- Multinational production transmits technology internationally by allowing high-productivity firms to produce in host countries.

9 Conclusion

9.1 Core conclusion

Tintelnot provides a quantitative theory of multinational production in which firms choose sparse global plant networks and use affiliates as export platforms. The paper shows that fixed costs and export platforms are both crucial for understanding the geography of multinational production.

9.2 Economic intuition

A multinational plant is valuable not only because it serves the host country, but because it can serve many nearby or low-cost destination markets. Opening a plant requires a fixed cost, so only sufficiently productive firms operate multi-country production networks. Once plants are open, firm-product-specific productivity draws determine which plant serves which market.

9.3 Policy implication

Trade and investment agreements affect third countries through the relocation of multinational production. Standard trade models and multinational models without export-platform sales can miss these effects.

9.4 Takeaway

Global production is not just trade plus FDI. It is a joint location, sourcing, and technology-transfer problem. A realistic quantitative model must let multinational affiliates export and must include fixed costs that make production networks sparse.